ICSE Class X Chemistry

Chapter-wise Question Bank: Chapters 4–7

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How to use this: Attempt every question cold on paper. Mark only after finishing a section. Log each wrong or blank item as a gap, rewrite it, and retry it after 24 hours. Source tags: **T23–T26** = board paper of that year; **S27 / S26 / S24** = specimen paper of that year. Items without a tag come from the syllabus and are board-style.

Chapter 4: Analytical Chemistry

A. Theory

4.1 Write the colour of the salt and of its solution for compounds of Fe^{2+}, Fe^{3+}, Cu^{2+}, Zn^{2+}, Pb^{2+} and NH_4^+.

4.2 Describe what you observe when NaOH solution is added drop by drop and then in excess to solutions of $FeSO_4$, $FeCl_3$, $CuSO_4$, $ZnSO_4$ and $Pb(NO_3)_2$. Give the colour of each hydroxide precipitated and write the equation.

4.3 Repeat 4.2 with NH_4OH solution in place of NaOH.

4.4 State the special action of (a) NH_4OH on a solution of a copper salt, (b) NaOH on an ammonium salt. Give the test for ammonia gas.

4.5 Define amphoteric. Show with equations that Al, Zn and Pb, and their oxides and hydroxides, react with caustic alkali (NaOH or KOH).

4.6 Give one test for each anion: carbonate, sulphide, sulphite, chloride, nitrate (brown ring test) and sulphate.

4.7 Why is NH_4OH preferred to NaOH to distinguish Zn^{2+} from Pb^{2+}, and what does the behaviour in excess tell you?

B. Objective questions

4.8 ___ solution forms a coloured precipitate with ammonium hydroxide which is soluble in excess of ammonium hydroxide: ferrous chloride or copper nitrate? [T25]

4.9 The metallic hydroxide which forms a deep inky blue solution with excess ammonium hydroxide is: (a) $Fe(OH)_2$ (b) $Cu(OH)_2$ (c) $Ca(OH)_2$ (d) $Fe(OH)_3$ [T23]

4.10 ___ is a white precipitate that is soluble in excess ammonium hydroxide solution: silver chloride or lead chloride? [T23]

4.11 The metal hydroxide which reacts with both acids and alkalis to form salt and water: (a) calcium hydroxide (b) magnesium hydroxide (c) aluminium hydroxide (d) ferric hydroxide [T24]

4.12 Which metal oxide can react with an acid as well as an alkali: silver oxide, calcium oxide, copper(II) oxide or aluminium oxide? [S26]

4.13 From iron, magnesium sulphite, zinc, sodium sulphide, lead, ferric chloride, copper and ferrous sulphate, choose: (a) the metal that will not produce hydrogen with dilute acids (b) the compound that produces sulphur dioxide with dil. HCl (c) the compound whose solution gives a dirty green precipitate with NaOH. [T23]

4.14 From PbO, CH_4, PbO_2, CO_2, HCl, NCl_3, SO_2 choose: (a) a colourless gas that turns acidified $K_2Cr_2O_7$ from orange to green (b) a yellow explosive oily liquid (c) the yellow metallic oxide formed on thermal decomposition of $PbCO_3$. [T25]

C. Identification and distinguishing

4.15 Alex tests an unknown salt Y. With silver nitrate it gives a white precipitate soluble in NH_4OH. With a few drops of NaOH it gives a pale blue precipitate. Identify the anion and the cation. [T26]

4.16 Rita tests a salt: NH_4OH gives a reddish-brown precipitate and $AgNO_3$ gives a white precipitate. Name the cation and the anion. [T24]

4.17 Identify the cation: (a) NH_4OH gives a white precipitate that does not dissolve in excess (b) NaOH gives a white precipitate insoluble in excess. [T23]

4.18 Identify the anion: (a) the compound on heating gives a colourless, odourless gas that turns lime water milky and has no effect on acidified $K_2Cr_2O_7$ (b) the solution with conc. H_2SO_4 and freshly prepared $FeSO_4$ gives a brown ring. [T24]

4.19 Name the anion: (a) solid P warmed with conc. H_2SO_4 and copper turnings gives a reddish-brown gas (b) dil. H_2SO_4 added to salt R and heated gives a colourless gas that turns moist lead acetate paper silvery black (c) barium nitrate solution added to salt solution Q gives a white precipitate insoluble in HCl. [T23]

4.20 Test tube C has $AgNO_3$ solution and test tube D has $Pb(NO_3)_2$ solution. Dilute HCl is added to each. State (a) one common observation (b) what happens on adding excess NH_4OH to the product in C and in D. [T25]

4.21 A chloride of iron R, treated with NaOH, gives products P and Q. Q is a reddish-brown precipitate. (a) Identify R. (b) Give the formulae of P and Q. (c) What do you observe when excess NaOH is added to Q? [S27]

4.22 A white salt X on strong heating gives a yellow residue, a colourless gas and a reddish-brown gas. Its solution with excess NH_4OH gives a chalky white insoluble precipitate. (a) Name the coloured gas. (b) Which cation is present? (c) Identify X. [S26]

4.23 Give reasons: (a) Can zinc and aluminium powders be distinguished by heating with concentrated NaOH solution? (b) Can zinc nitrate and lead nitrate be distinguished by adding NH_4OH solution? [S27, S26]

4.24 Can calcium nitrate and lead nitrate be distinguished by adding NH_4OH solution? Give the reason. [S26, S24]

4.25 Distinguish between: (a) ammonium chloride and sodium chloride using an alkali (b) zinc nitrate solution and calcium nitrate solution using excess NaOH solution. [T25]

4.26 Give one observation each: (a) an amphoteric metal is added to hot concentrated alkali [T26] (b) NaOH solution is added to copper nitrate solution, first a few drops and then in excess [T23]

4.27 Name the anion present in a salt that produces a gas with the smell of rotten eggs when treated with dil. HCl. [S24]

Chapter 5: Mole Concept and Stoichiometry

A. Theory and definitions

5.1 State Avogadro's law and Gay-Lussac's law of combining volumes. Define molar volume. [T25, T23, T24]

5.2 Define mole, gram atomic mass, gram molecular mass and vapour density. State the relation between molecular mass and vapour density. [T24, T23]

5.3 Define empirical formula and molecular formula. State the relation between them. [T24]

5.4 State the relations between mass, moles, volume at STP and number of molecules (Avogadro's number).

B. Objective questions

5.5 What volume of CO_2 is produced at STP when 5 litres of propane is burnt completely? $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$. (a) 10 L (b) 15 L (c) 20 L (d) 25 L [T26]

5.6 40 cm^3 of methane reacts with 60 cm^3 of oxygen: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. All volumes at room temperature. Total volume of gases remaining: (a) 60 cm^3 (b) 40 cm^3 (c) 45 cm^3 (d) 50 cm^3 [T25]

5.7 An organic compound has vapour density 22. Its molecular formula is: (a) CH_4 (b) C_2H_4 (c) C_2H_6 (d) C_3H_8 [T25]

5.8 Which would occupy 22.4 litres at STP? 1. 32 g of oxygen 2. 2 moles of hydrogen 3. 6.022×10^{23} molecules of ammonia. (a) 1 and 2 (b) 1 and 3 (c) 2 and 3 (d) 1, 2 and 3 [T24]

5.9 The vapour density of CH_3OH is: (a) 32 (b) 18 (c) 16 (d) 34 [T23]

5.10 Volume occupied at STP by (a) 2 moles (b) 0.2 moles of a gas. [S26, S27]

5.11 The ratio of the volumes occupied by 22 g of carbon dioxide and 10 g of hydrogen gas is: (a) 2.2 : 1 (b) 1 : 2.2 (c) 1 : 10 (d) 10 : 1 [S26]

5.12 A compound of carbon and hydrogen only has vapour density 14. Its molecular formula is: (a) CH (b) C_2H_4 (c) C_2H_2 (d) CH_2 [S27]

5.13 Atomic masses of S, O and He are 32, 16 and 4. P: 16 g of oxygen has four times the number of atoms in 4 g of helium. Q: 16 g of oxygen has half the number of atoms in 32 g of sulphur. (a) only P (b) only Q (c) both P and Q (d) neither [S24]

5.14 The volume occupied by 8 grams of oxygen gas at STP is: 5.6 litres or 8.96 litres? [T26]

C. Numericals

5.15 Equal volumes of NH_3 and Cl_2 are kept at the same temperature and pressure. Find the number of molecules in the chlorine gas when the mass of ammonia is 34 g. (Cl = 35.5, H = 1, N = 14) [T26]

5.16 $Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2$. 32.5 g of zinc reacts. (a) Moles of zinc (b) mass of HNO_3 needed (c) volume of NO_2 liberated at STP. (H = 1, N = 14, O = 16, Zn = 65) [T26]

5.17 5.6 litres of gas Z weighs 32 g at STP. Find its molecular weight and vapour density. [T26]

5.18 $Mg + 2H_2SO_4 \rightarrow MgSO_4 + 2H_2O + SO_2$. 60 g of magnesium is used. (a) Mass of H_2SO_4 needed (b) volume of SO_2 liberated at STP. (Mg = 24, H = 1, S = 32, O = 16) [T25]

5.19 A gas cylinder can hold 150 g of hydrogen at certain temperature and pressure. An identical cylinder holds 450 g of gas G under the same conditions. Find (a) the vapour density of G (b) the molecular weight of G. [T25]

5.20 An organic compound X contains C, H and O only. C = 47.4%, H = 10.5%, relative molecular mass 76. Find its empirical and molecular formula. (C = 12, O = 16, H = 1) [T25]

5.21 $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. (RMM of $CaCO_3$ = 100) (a) Mass of 5 moles of $CaCO_3$ (b) moles of HCl that react with it (c) volume of CO_2 at STP at the same time. [T24]

5.22 $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$. 240 cc of ammonia is burnt in 300 cc of oxygen. Find the composition of the resultant gaseous mixture at room temperature. [T24]

5.23 $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. 80 ml of ethane is burnt in 300 ml of oxygen. Find the composition of the resultant gaseous mixture at room temperature. [T23]

5.24 The empirical formula of an organic compound is $CHCl_2$ and its relative molecular mass is 168. Find its molecular formula. (C = 12, H = 1, Cl = 35.5) [T23]

5.25 (a) Find the percentage of phosphorus in the fertilizer superphosphate $Ca(H_2PO_4)_2$, correct to one decimal place. (H = 1, P = 31, O = 16, Ca = 40) (b) Write the empirical formula of C_8H_{18}. [T23]

5.26 The empirical formula of an organic compound is C_3H_4N and its molecular weight is 108. Find the amount of carbon in one mole of the compound. Show all steps. (C = 12, H = 1, N = 14) [S24]

5.27 A gas cylinder of capacity 40 dm^3 is filled with gas X of mass 20 g. The same cylinder filled with hydrogen at the same temperature and pressure holds 2 g. Find the relative molecular mass of X. [S24]

5.28 8.2 g of calcium nitrate is heated: $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$. Find (a) the volume of NO_2 at STP (b) the mass of CaO formed. (Ca = 40, N = 14, O = 16) [S24]

5.29 Calculate the number of moles in 22 grams of carbon dioxide. [S24]

5.30 What mass of carbon dioxide contains the same number of molecules as 3.2 g of oxygen gas? (O = 16, C = 12) [S26]

5.31 20 g of gas M_2P has fewer moles than 20 g of gas MP. Justify with calculations. (M = 14, P = 16) [S27]

5.32 $(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + 4H_2O$. 126 g of ammonium dichromate decomposes. Calculate (a) moles decomposed (b) mass of Cr_2O_3 formed (c) volume of N_2 at STP. (N = 14, Cr = 52, O = 16, H = 1) [S27, S26]

5.33 What is the ratio of methane to the acidic gaseous product when methane burns completely? Give the balanced equation. [S27]

Chapter 6: Electrolysis

A. Theory

6.1 Define electrolyte and non-electrolyte, strong and weak electrolyte, with two examples each. Classify substances as containing molecules only, ions only, or both molecules and ions.

6.2 Define electrolysis, electrode, anode, cathode, anion, cation, oxidation and reduction (as loss and gain of electrons). [T24, T23]

6.3 Why is the anode called the oxidising electrode? [T25]

6.4 What is selective discharge? State the factors that decide which ion is discharged, with examples from the electrochemical series.

6.5 Electrolysis of molten lead bromide: name the electrodes, observations, and anode and cathode reactions. Why are graphite electrodes preferred? [T25]

6.6 Electrolysis of acidified water with platinum electrodes: why is it acidified, what are the cathode and anode reactions, and what is the ratio of gases by volume? [T25, S27]

6.7 Electrolysis of aqueous $CuSO_4$ with (a) platinum electrodes (b) copper electrodes: give observations at each electrode, the change in colour of the electrolyte and the equations.

6.8 Electroplating: state the reasons and conditions. Name the electrolyte and electrodes for nickel plating and silver plating, and give electrode equations. [T26, T24, S26]

6.9 Electrorefining of copper: name the anode, cathode and electrolyte, and give the equations. [T23]

B. Objective questions

6.10 In the electrolysis of aqueous copper(II) sulphate with copper electrodes: (a) copper is deposited at the negative electrode (b) oxygen is produced at the positive electrode (c) the positive electrode increases in mass (d) the negative electrode decreases in mass. Which is correct? [T26]

6.11 For the same electrolysis, which statement is NOT correct? (a) an oxidation reaction occurs at the positive electrode (b) current is carried by ions (c) the positive electrode loses mass (d) the number of Cu^{2+} ions in the electrolyte decreases [T25]

6.12 Assertion (A): In the electrolysis of acidified water, the volume of hydrogen liberated is twice the volume of oxygen. Reason (R): Water has hydrogen and oxygen in the ratio 1 : 2 by volume. (a) A true, R false (b) A false, R true (c) both true, R explains A (d) both true, R does not explain A [T26]

6.13 In the electrolysis of molten Al_2O_3: 1. $Al^{3+} + 3e^- \rightarrow Al$ 2. $2O^{2-} \rightarrow O_2 + 4e^-$ 3. $C + O_2 \rightarrow CO_2$ 4. $Al_2O_3 \rightarrow 2Al^{3+} + 3O^{2-}$. Which takes place at the negative electrode? [T26]

6.14 Aqueous copper sulphate turns colourless on electrolysis. Which could be the electrodes? P: anode copper, cathode copper. Q: anode platinum, cathode copper. R: anode copper, cathode platinum. (a) only P (b) only Q (c) only R (d) Q and R [S24]

6.15 During electrolysis of copper sulphate solution, if ___ is used as electrodes the colour of the electrolyte does not fade: copper or platinum? [T24]

6.16 Which reaction takes place at the anode during silver electroplating? (a) $Ag - 1e^- \rightarrow Ag^+$ (b) $Ag + 1e^- \rightarrow Ag^-$ (c) $Ag - 1e^- \rightarrow Ag$ (d) none of the above [T23]

6.17 Fill in: (a) During electrolysis the compound ___ in molten state liberates reddish-brown fumes at the anode: NaCl or $PbBr_2$. (b) The ion discharged most readily is ___: Fe^{2+} or Cu^{2+}. [T23]

6.18 Arrange Cu^{2+}, Na^{+}, Zn^{2+}, Ag^{+} in the order of preferential discharge at the cathode. [T23]

6.19 The reaction at the anode during the electrolysis of molten lead bromide is: $2Br - 2e^{-} \rightarrow 2Br^{-}$ or $2Br^{-} - 2e^{-} \rightarrow Br_2$? [T26]

6.20 Which particles conduct electricity in dilute H_2SO_4 and in the compound XSO_4? (a) electrons (b) only positive ions (c) only negative ions (d) both positive and negative ions [S26]

6.21 Which statement about the flow of current is correct? (a) In all aqueous solutions current is carried by electrons. (b) In all acidic solutions current is carried by ions. (c) In molten electrolytes current is carried by electrons. (d) In metal wires current is carried by ions. [S27]

6.22 In electrorefining of copper, which is correct? (a) the anode is pure copper (b) the cathode is impure copper (c) copper is deposited at the anode (d) Cu ions from the anode move to the cathode and deposit as pure copper [S26]

6.23 Match: (a) $CuO + C \rightarrow Cu + CO$ (b) $Al^{3+} + 3e^{-} \rightarrow Al$ (c) $PbBr_2 \rightarrow Pb^{2+} + 2Br^{-}$ (d) $2O^{2-} - 4e^{-} \rightarrow O_2$ (e) $HCl \rightarrow H^{+} + Cl^{-}$ with: oxidation, ionisation, reduction, redox, electrolytic dissociation. [S27, T26]

6.24 X: $HCl \rightleftharpoons H^{+} + Cl^{-}$ (in solution). Y: $PbBr_2 \rightleftharpoons Pb^{2+} + 2Br^{-}$ (in molten state). Which reaction shows (a) electrolytic dissociation (b) ionisation? [T24]

C. Application

6.25 A spoon is electroplated with silver. (a) Write the cathode reaction. (b) Why is silver nitrate solution not preferred as electrolyte? (c) Why is alternating current not used? (d) What is the observation at the anode? [T26]

6.26 An iron cup is electroplated with nickel. (a) Why is the cup placed at the cathode? (b) Name the ion that must be present in the electrolyte. (c) State one condition for a smooth, firm and even deposit. (d) Write the cathode reaction. (e) What change is seen at the anode? [T24]

6.27 Silver plating a steel object involves three steps: copper, then nickel, then silver. (a) In step 1, what is seen on the object's surface and why does the Cu^{2+} concentration stay constant? (b) State two changes needed to coat nickel in step 2. (c) Write the reaction at the positive electrode in step 3. [S26]

6.28 Write the reaction at the anode during electrolysis of aqueous sodium argentocyanide with silver electrodes. [T25]

6.29 Acidulated water is electrolysed using platinum electrodes. (a) Why is dilute sulphuric acid added? (b) Write the cathode reaction. (c) What is the observation at the anode? [T25]

6.30 Copper sulphate solution is electrolysed using copper electrodes. (a) Which electrode is the oxidising electrode and why? (b) Write the equation at that electrode. [S24]

6.31 With reference to electrorefining of copper: (a) what is the anode made of? (b) what do you observe at the cathode? (c) write the cathode reaction. [T23]

6.32 In the electrolysis of acidulated water (A is the anode, B the cathode): (a) write the equation at A (b) give a test for the product at A (c) which gas is liberated at B? (d) why is water acidified? (e) what is the ratio by volume of the gases at A and B? [S27]

6.33 Why is silver-plated cutlery not considered pure silver? [S26]

Chapter 7: Metallurgy

A. Theory

7.1 Define mineral, ore, gangue (matrix), flux and slag. Name the chief ores of iron, aluminium and zinc with formulae.

7.2 Describe the stages in the extraction of a metal from its ore.

7.3 Explain hydraulic washing, magnetic separation and froth flotation. For which ores is each used?

7.4 Define roasting and calcination. Give one equation for each.

7.5 Which oxides can be reduced by H_2, C or CO (CuO, PbO, Fe_2O_3, ZnO)? Why can Al_2O_3 and MgO not be reduced this way? How are sodium, potassium and calcium obtained?

7.6 Baeyer's process: what is the purpose of NaOH? Write the equation for bauxite with NaOH and the conversion of $Al(OH)_3$ to alumina. [T26, T25]

7.7 Hall-Héroult process: composition of the electrolyte, role of cryolite and fluorspar, electrodes, electrode reactions, why the carbon anode needs periodic replacement, and why coke is sprinkled on top of the electrolyte.

7.8 Define alloy and amalgam. Give the composition and one use of stainless steel, magnalium, duralumin, brass, bronze and fuse metal (solder). [T23, S24, S26]

B. Objective questions

7.9 The ore that can be concentrated by magnetic separation: corundum, haematite, calamine or bauxite? [T25]

7.10 From zinc blende, C_2H_2, calamine, CH, haematite choose: (a) the ore concentrated by magnetic separation (b) the empirical formula of ethyne. [T24]

7.11 The metal whose oxide can be reduced by common reducing agents: copper, sodium, aluminium or potassium? [T25]

7.12 The metallic oxide which can be reduced by common reducing agents: Fe_2O_3 or Al_2O_3? [T24]

7.13 Which compound is NOT an ore of aluminium: cryolite, corundum, fluorspar or bauxite? [T23]

7.14 A mineral from which the metal can be extracted economically and conveniently is: matrix, ore, flux or alloy? [T24]

7.15 The process of heating the concentrated ore in a limited supply of air or in its absence: roasting or calcination? [T24, S27]

7.16 Zinc blende is converted to zinc oxide by: calcination or roasting? [T25]

7.17 Which compound is NOT a constituent of the electrolytic mixture in Hall-Héroult's process: Al_2O_3, $NaAlO_2$, Na_3AlF_6 or CaF_2? [T24]

7.18 Copper, zinc and tin are alloyed to form: duralumin, brass, bronze or solder? [T24]

7.19 Name the main metal in (a) duralumin (b) stainless steel. [T24]

7.20 Identify the alloy: (a) aluminium, magnesium, manganese, copper (b) iron, nickel, chromium, carbon. [T23]

7.21 Name the alloy made of (a) copper, zinc and tin (b) lead and tin. [S24]

7.22 Name the main metal in duralumin and in brass. Which alloy is made of aluminium and magnesium only? [S26, S27]

7.23 The alloy that contains a non-metal as one of its constituents: bronze or steel? [S27]

7.24 Name the non-metallic element used as the anode during electrolytic reduction of fused alumina in Hall-Héroult's process. [S27]

C. Application

7.25 In Bayer's process: (a) why is caustic soda added? (b) write the balanced equation between bauxite and caustic soda. [T26]

7.26 With reference to the concentration of bauxite: (a) name the process used (b) give the balanced equation for conversion of aluminium hydroxide to pure alumina. [T25]

7.27 In the Hall-Héroult cell diagram: (a) name the electrolyte constituent that has a divalent metal (b) name the powder X sprinkled on the surface of the electrolyte (c) name the process (d) write the reactions at anode Y and cathode Z. [T23]

7.28 (a) Name the other aluminium-containing compound added to alumina. (b) Give the equation for the reaction at the cathode. [S26, S24]

7.29 Write one use of bronze and one use of fuse metal. [T23]

7.30 Write balanced equations for: roasting of ZnS, calcination of $ZnCO_3$, reduction of ZnO with carbon, reduction of Fe_2O_3 with CO, reduction of CuO with H_2, reduction of PbO with carbon.

7.31 Why is alumina electrolysed in molten cryolite and not as pure alumina?

Answer Key

Objective and numerical items only. Write theory items from your own notes, then check them against your textbook.

Chapter 4

4.8 copper nitrate

4.9 (b) $Cu(OH)_2$

4.10 silver chloride

4.11 (c) aluminium hydroxide

4.12 aluminium oxide

4.13 (a) copper (b) magnesium sulphite (c) ferrous sulphate

4.14 (a) SO_2 (b) NCl_3 (c) PbO

4.15 anion: chloride (Cl^-); cation: copper (Cu^{2+})

4.16 cation: Fe^{3+}; anion: Cl^-

4.17 (a) Pb^{2+} (b) Ca^{2+}

4.18 (a) carbonate (b) nitrate

4.19 (a) nitrate (b) sulphide (c) sulphate

4.20 (a) a white precipitate forms in both (b) in C the precipitate dissolves to give a clear solution; in D it stays insoluble

4.21 (a) $FeCl_3$ (b) P = NaCl, Q = $Fe(OH)_3$ (c) the precipitate stays insoluble in excess

4.22 (a) nitrogen dioxide (b) Pb^{2+} (c) lead nitrate, $Pb(NO_3)_2$

4.23 (a) No, both liberate hydrogen which burns with a pop (b) Yes, both give a white precipitate but $Zn(OH)_2$ dissolves in excess NH_4OH while $Pb(OH)_2$ does not

4.24 Yes: lead nitrate gives a white precipitate insoluble in excess; calcium nitrate gives no precipitate (no visible reaction)

4.25 (a) warm with NaOH: NH_4Cl gives pungent ammonia, NaCl gives nothing (b) Zn nitrate: white precipitate dissolving in excess NaOH; Ca nitrate: white precipitate not dissolving in excess

4.26 (a) a colourless gas (hydrogen) evolves and burns with a pop (b) pale blue precipitate, insoluble in excess NaOH

4.27 sulphide

Chapter 5

5.5 (b) 15 L

5.6 (b) 40 cm^3. Oxygen is limiting: 30 cm^3 CH_4 reacts, 10 cm^3 CH_4 is left, 30 cm^3 CO_2 forms; 10 + 30 = 40 cm^3 (water is liquid)

5.7 (d) C_3H_8 (molecular mass 44)

5.8 (b) 1 and 3

5.9 (c) 16

5.10 (a) 44.8 L (b) 4.48 L

5.11 (c) 1 : 10 (0.5 mol : 5 mol)

5.12 (b) C_2H_4 (molecular mass 28)

5.13 (d) neither (each sample is 1 mole of atoms)

5.14 5.6 litres

5.15 34 g NH_3 = 2 mol, so Cl_2 has 2 mol = 1.2044×10^{24} molecules

5.16 (a) 0.5 mol (b) 126 g (c) 22.4 L (0.5 mol Zn gives 1 mol NO_2)

5.17 molecular weight 128; vapour density 64

5.18 (a) 490 g (b) 56 L

5.19 same volume means same moles: 150 g H_2 = 75 mol, so 450 g G = 75 mol; molecular weight 6; vapour density 3

5.20 O% = 42.1; ratio C : H : O = 3 : 8 : 2; empirical formula $C_3H_8O_2$ (mass 76); molecular formula $C_3H_8O_2$

5.21 (a) 500 g (b) 10 mol (c) 112 L

5.22 N_2 120 cc and O_2 120 cc remain (180 cc O_2 used; water is liquid)

5.23 CO_2 160 ml and O_2 20 ml remain

5.24 empirical mass 84, n = 2, so $C_2H_2Cl_4$

5.25 (a) 234 g/mol, so P = 62/234 = 26.5% (b) C_4H_9

5.26 empirical mass 54, n = 2, formula $C_6H_8N_2$; carbon = 72 g

5.27 20

5.28 (a) 2.24 L (b) 2.8 g

5.29 0.5 mol

5.30 4.4 g

5.31 M_2P = 44 g/mol, so 20 g = 0.45 mol; MP = 30 g/mol, so 20 g = 0.67 mol

5.32 (a) 0.5 mol (b) 76 g (c) 11.2 L

5.33 1 : 1; $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Chapter 6

6.10 (a)

6.11 (d)

6.12 (a) A true, R false

6.13 1

6.14 (b) only Q

6.15 copper

6.16 (a) $Ag - 1e^- \rightarrow Ag^+$

6.17 (a) $PbBr_2$ (b) Cu^{2+}

6.18 $Ag^+ > Cu^{2+} > Zn^{2+} > Na^+$

6.19 $2Br^- - 2e^- \rightarrow Br_2$

6.20 (d) both positive and negative ions

6.21 (b)

6.22 (d)

6.23 (a) redox (b) reduction (c) electrolytic dissociation (d) oxidation (e) ionisation

6.24 (a) Y (b) X

6.25 (a) $Ag^+ + e^- \rightarrow Ag$ (b) $AgNO_3$ gives a rough, non-adherent deposit; sodium argentocyanide gives a smooth, even coat (c) AC reverses polarity, so no steady deposit forms (d) the silver anode dissolves and loses mass

6.26 (a) the object to be plated is always the cathode (b) Ni^{2+} (c) clean surface, suitable electrolyte and low current (d) $Ni^{2+} + 2e^- \rightarrow Ni$ (e) the nickel anode dissolves

6.27 (a) reddish-brown (pink) deposit; the copper anode dissolves, replacing Cu^{2+} at the same rate it is deposited (b) the anode must be nickel and the electrolyte a nickel salt solution (c) $Ag \rightarrow Ag^+ + e^-$

6.28 $Ag \rightarrow Ag^+ + e^-$

6.29 (a) pure water is a poor conductor; the acid supplies ions (b) $2H^+ + 2e^- \rightarrow H_2$ (c) bubbles of oxygen

6.30 (a) the anode, because it loses electrons (b) $Cu - 2e^- \rightarrow Cu^{2+}$

6.31 (a) impure copper (b) pure copper is deposited (c) $Cu^{2+} + 2e^- \rightarrow Cu$

6.32 (a) $4OH^- - 4e^- \rightarrow 2H_2O + O_2$ (b) the gas rekindles a glowing splinter (c) hydrogen (d) pure water does not conduct; the acid provides ions (e) A : B = 1 : 2

6.33 only a thin outer coating is silver; the base metal lies underneath

Chapter 7

7.9 haematite

7.10 (a) haematite (b) CH

7.11 copper

7.12 Fe_2O_3

7.13 fluorspar

7.14 ore

7.15 calcination

7.16 roasting

7.17 $NaAlO_2$

7.18 bronze

7.19 (a) aluminium (b) iron

7.20 (a) duralumin (b) stainless steel

7.21 (a) bronze (b) solder (fuse metal)

7.22 aluminium; copper; magnalium

7.23 steel (contains carbon)

7.24 carbon (graphite)

7.25 (a) NaOH dissolves the amphoteric alumina as sodium aluminate and leaves iron oxide and other impurities undissolved (b) $Al_2O_3 \cdot 2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$

7.26 (a) Baeyer's process (b) $2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$ (on heating)

7.27 (a) calcium fluoride (fluorspar) (b) coke powder (c) Hall-Héroult process (d) anode: $2O^{2-} \rightarrow O_2 + 4e^-$ (the O_2 burns the carbon anode to CO_2); cathode: $Al^{3+} + 3e^- \rightarrow Al$

7.28 (a) cryolite, Na_3AlF_6 (b) $Al^{3+} + 3e^- \rightarrow Al$

7.29 bronze: statues, medals, utensils; fuse metal: fuse wire, soldering

7.30 $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; $ZnCO_3 \rightarrow ZnO + CO_2$; $ZnO + C \rightarrow Zn + CO$; $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$; $CuO + H_2 \rightarrow Cu + H_2O$; $PbO + C \rightarrow Pb + CO$

7.31 pure alumina melts near 2050 °C and conducts poorly; cryolite lowers the melting point to about 950 °C and improves conductivity

Next banks: Chapter 8 (Study of Compounds) and Chapter 9 (Organic Chemistry) follow once you have cleared and logged gaps from this set.